

ASME

Digital Hybrid Piano

Electrical Overview

EOH 2026

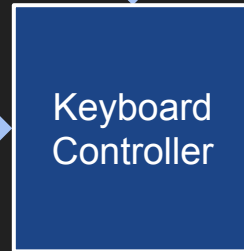
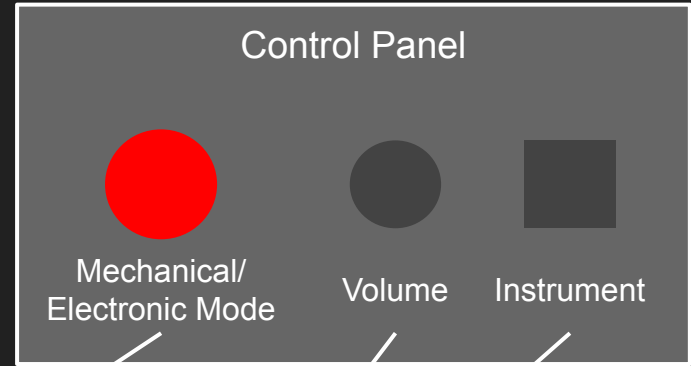
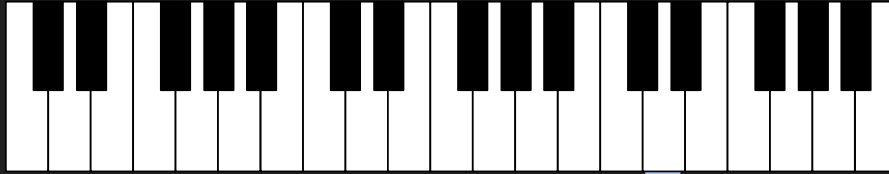
Goals for the electrical system

- Read key presses from the piano
 - Velocity sensitive (volume of note is based on how hard you press a key)
 - Polyphonic (handle/play multiple keys/notes at once)
- Turn key press information into sound
 - Piano note samples?
 - MIDI controller?
- Mechanically simple for ease of assembly
- Robust to withstand kids pounding on the piano during EOH

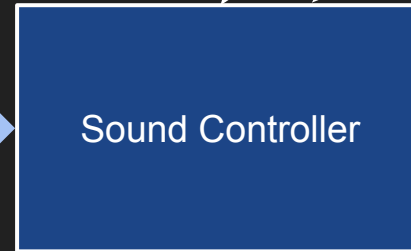


Electronics!

Electrical system



MIDI
Commands



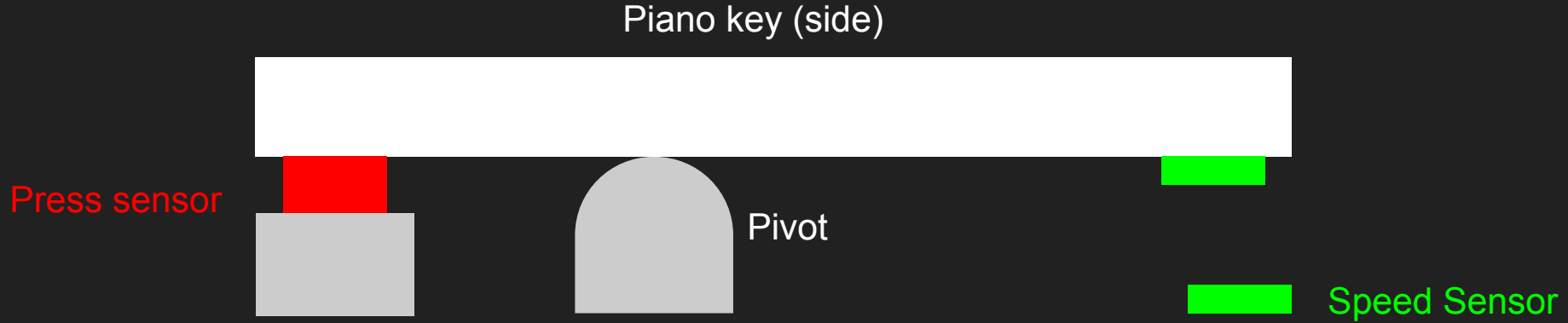
Keyboard Controller

The next section goes over the basics of the different components of the keyboard controller:

- Velocity-sensitive key switch
- Arduino
- MIDI
- Shift registers

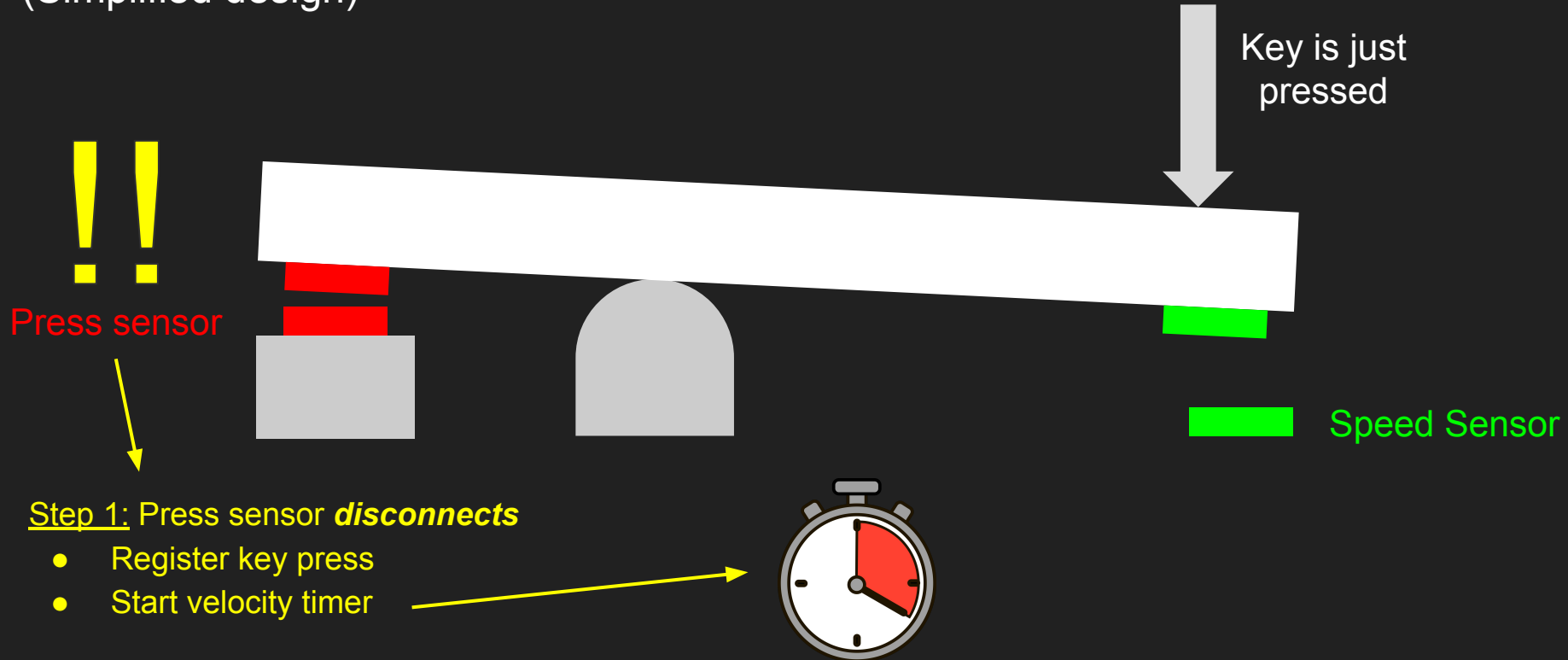
Velocity-sensitive key switch mechanism

(Simplified design)



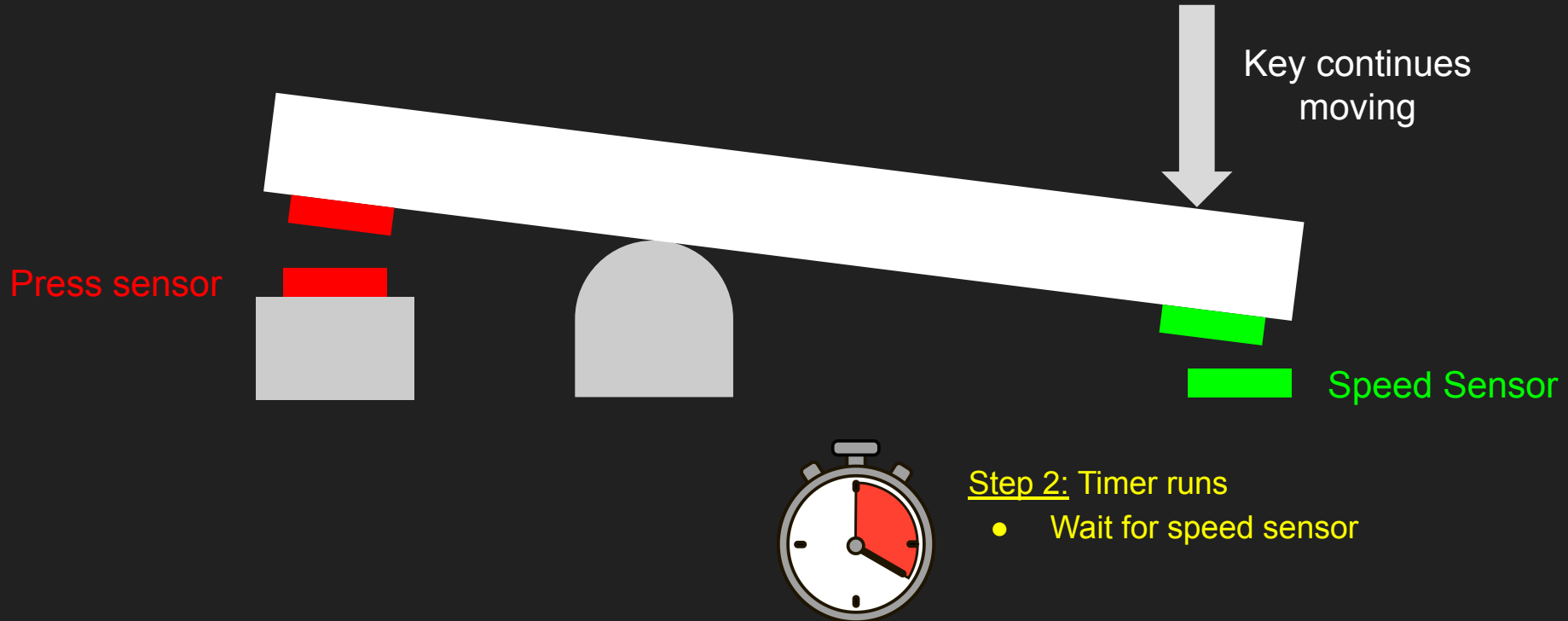
Velocity-sensitive key switch mechanism

(Simplified design)



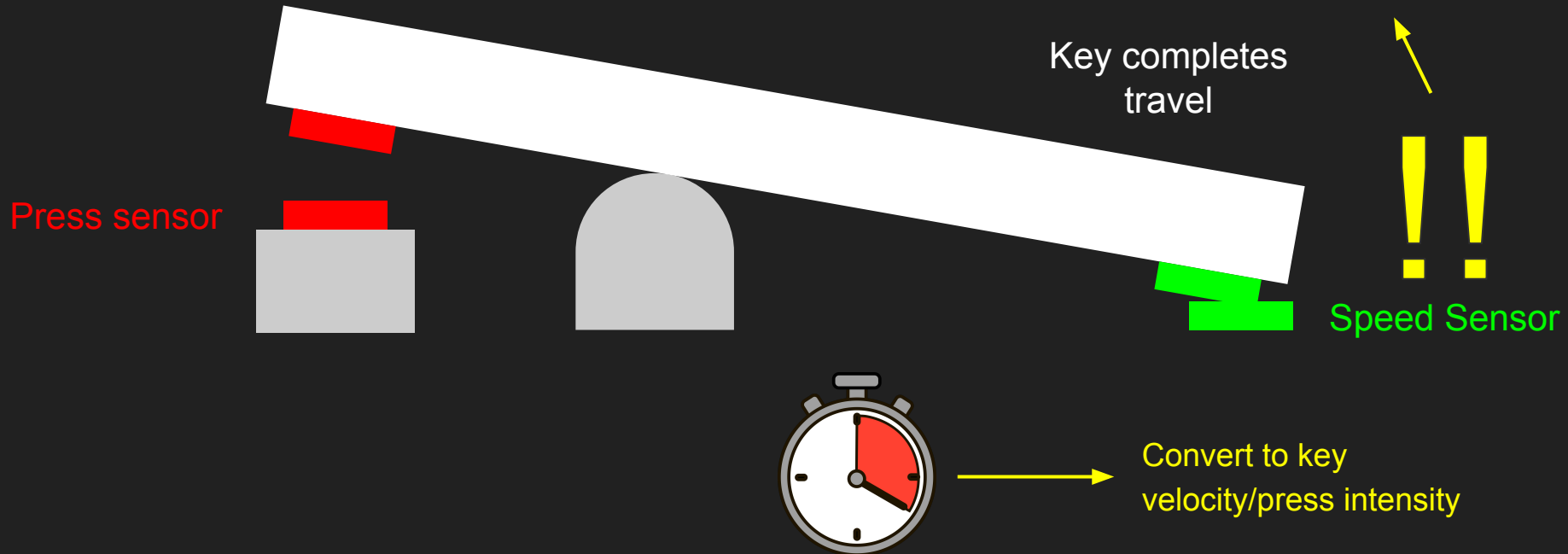
Velocity-sensitive key switch mechanism

(Simplified design)

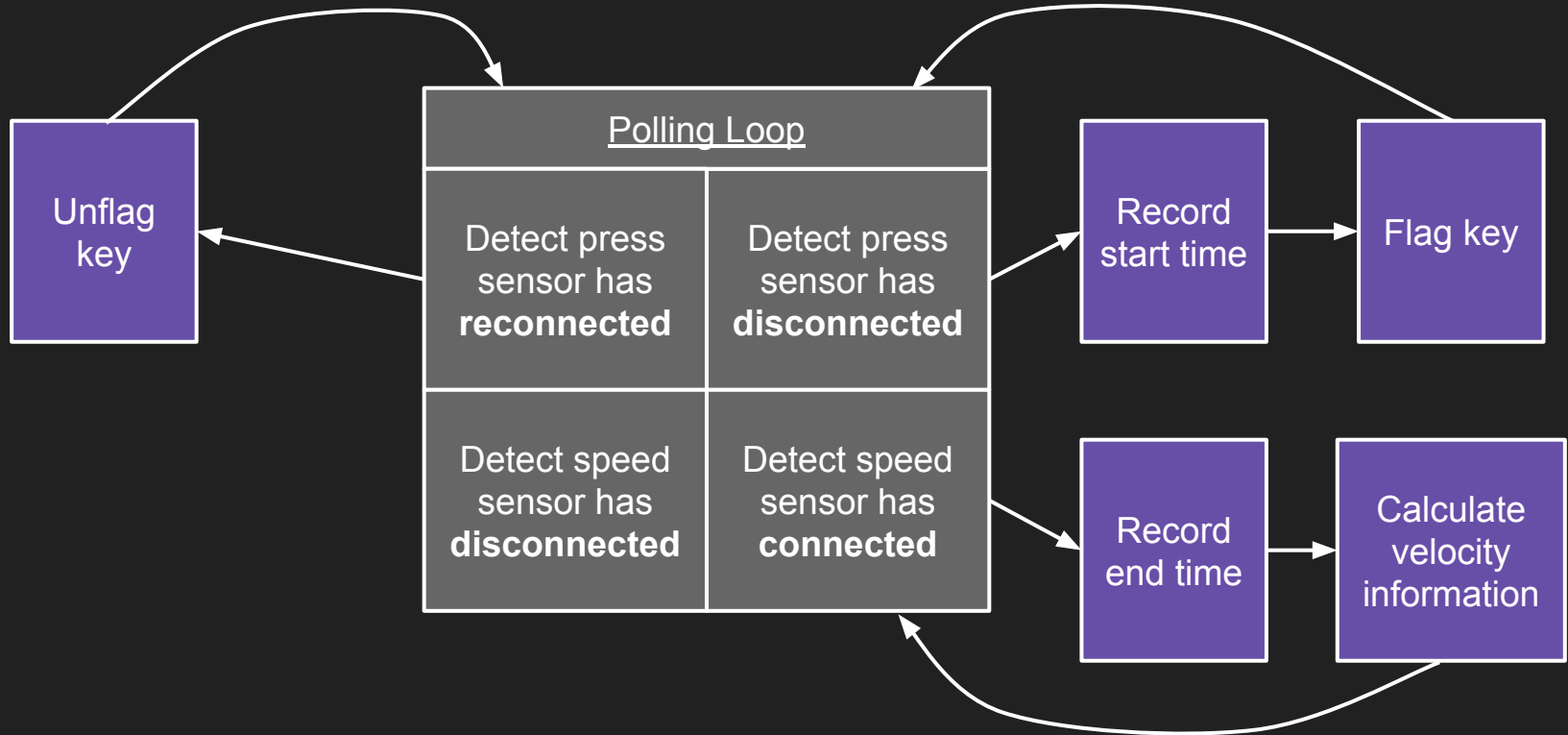


Velocity-sensitive key switch mechanism

(Simplified design)



Key sensing program flow

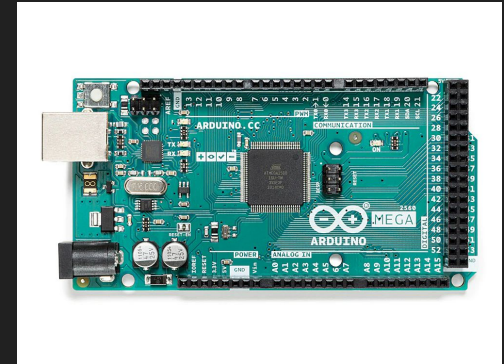


What is an Arduino?

- Arduino is a *microcontroller* development kit that provides hardware and software abstractions to make embedded systems development easy and accessible
- A kind of basic computer (8-bit) that is able to interact with electronic inputs/outputs (I/O)
- Able to execute basic programs but not really enough processing power for digital sound or images



Arduino UNO
(14 digital I/O)



Arduino MEGA
(54 digital I/O)

MIDI

- Musical Instrument Digital Interface
- Popular protocol that provides a standard language for communicating with digital instruments
- Each key, knob, slider, etc. interaction is converted to a unique MIDI event
- Software like digital audio workstations (DAWs) know how to interpret these commands and map them to sounds



MIDI messages

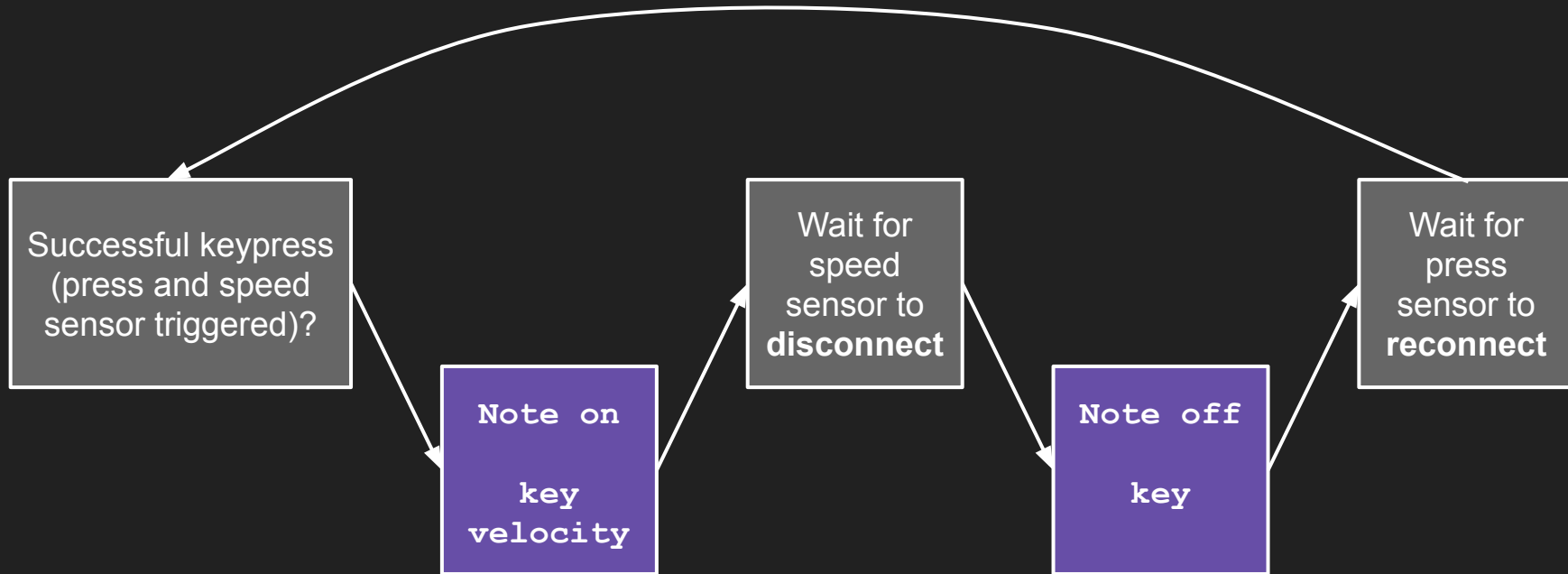
- Note on 0x9c 0xkk 0xvv
- Note off 0x8c 0xkk 0xvv

c: channel (0-15)

kk: key number (0-127)

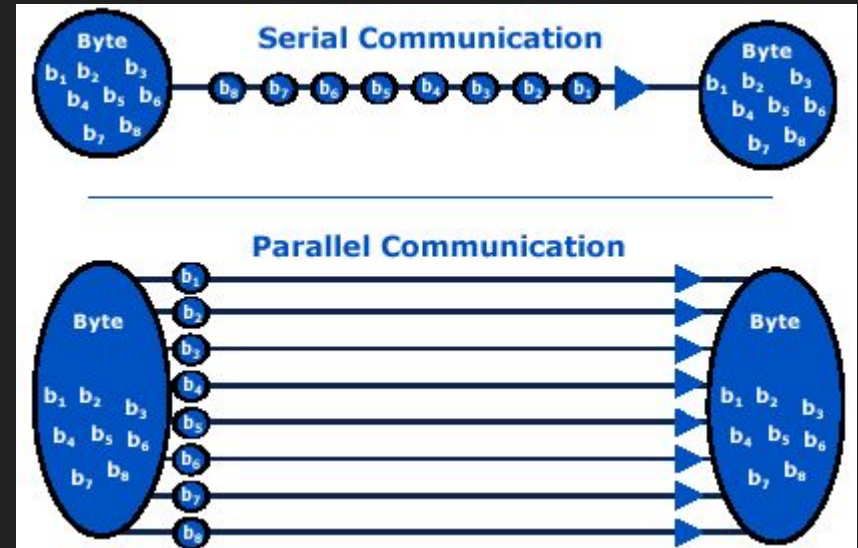
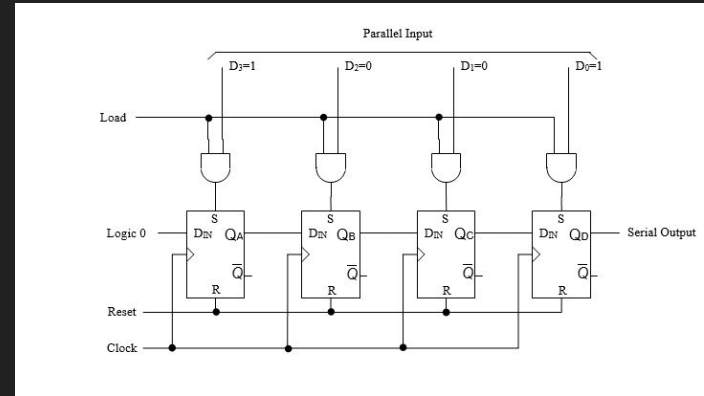
vv: striking velocity (0-127)

MIDI program flow

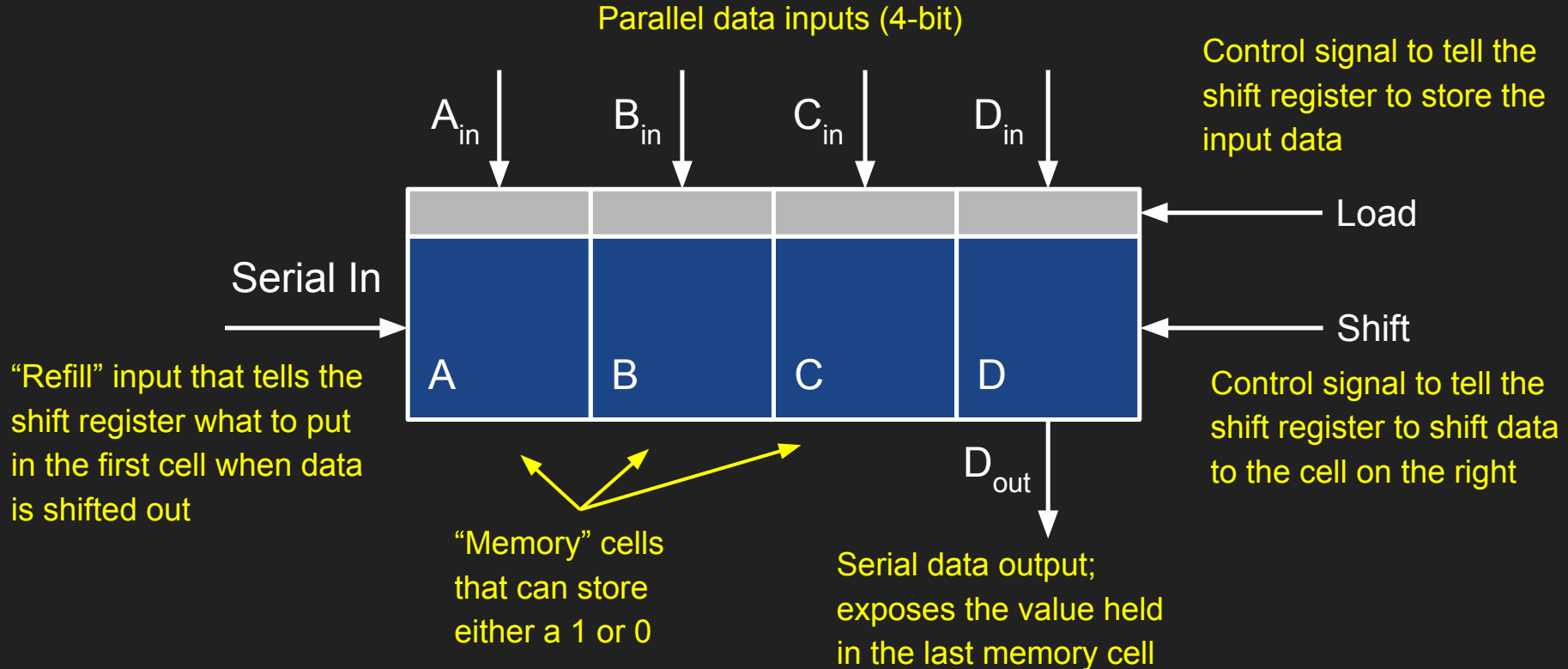


Shift registers

- A *shift register* is building block of digital circuits that stores digital data and can shift it from one location to another
- For this project we are interested in using a parallel-in serial-out (PISO) shift register as a *parallel to serial converter*
- Used to read all 176 key sensors with only three wires

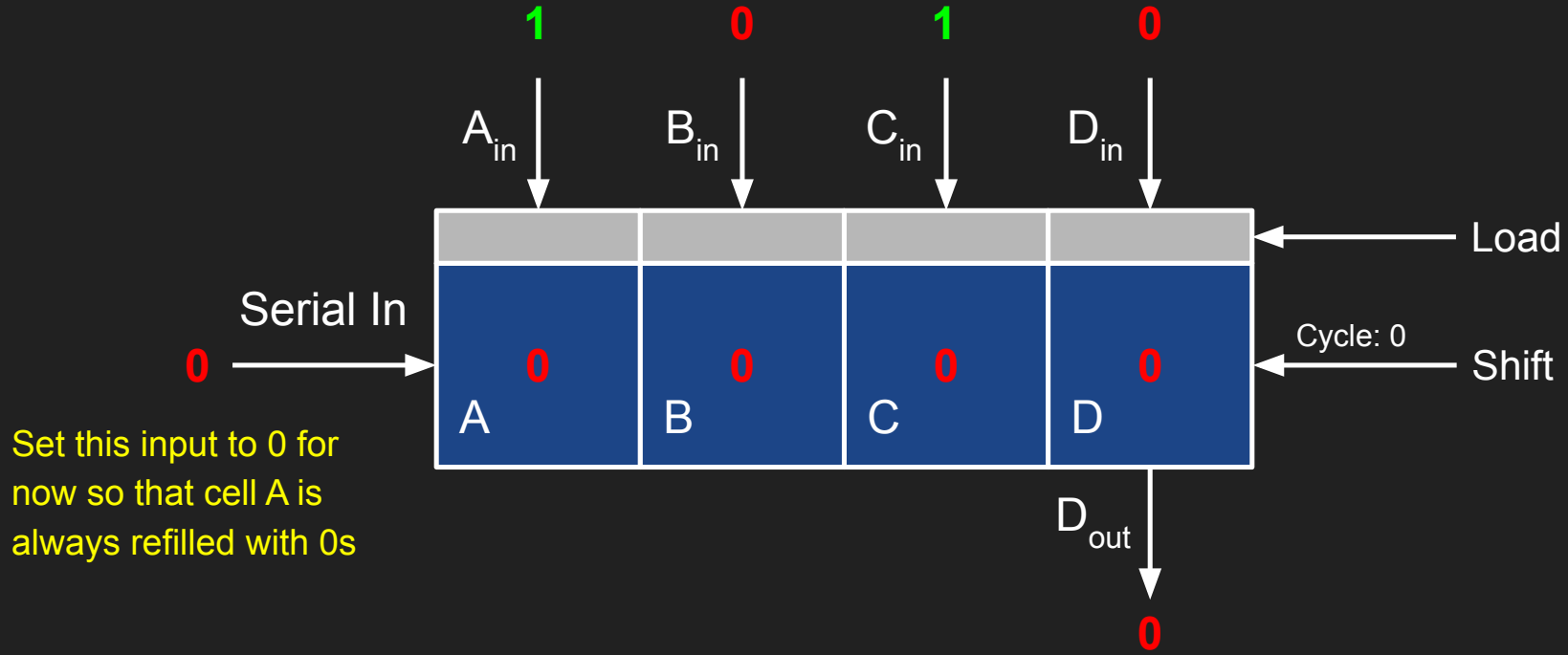


Shift register example

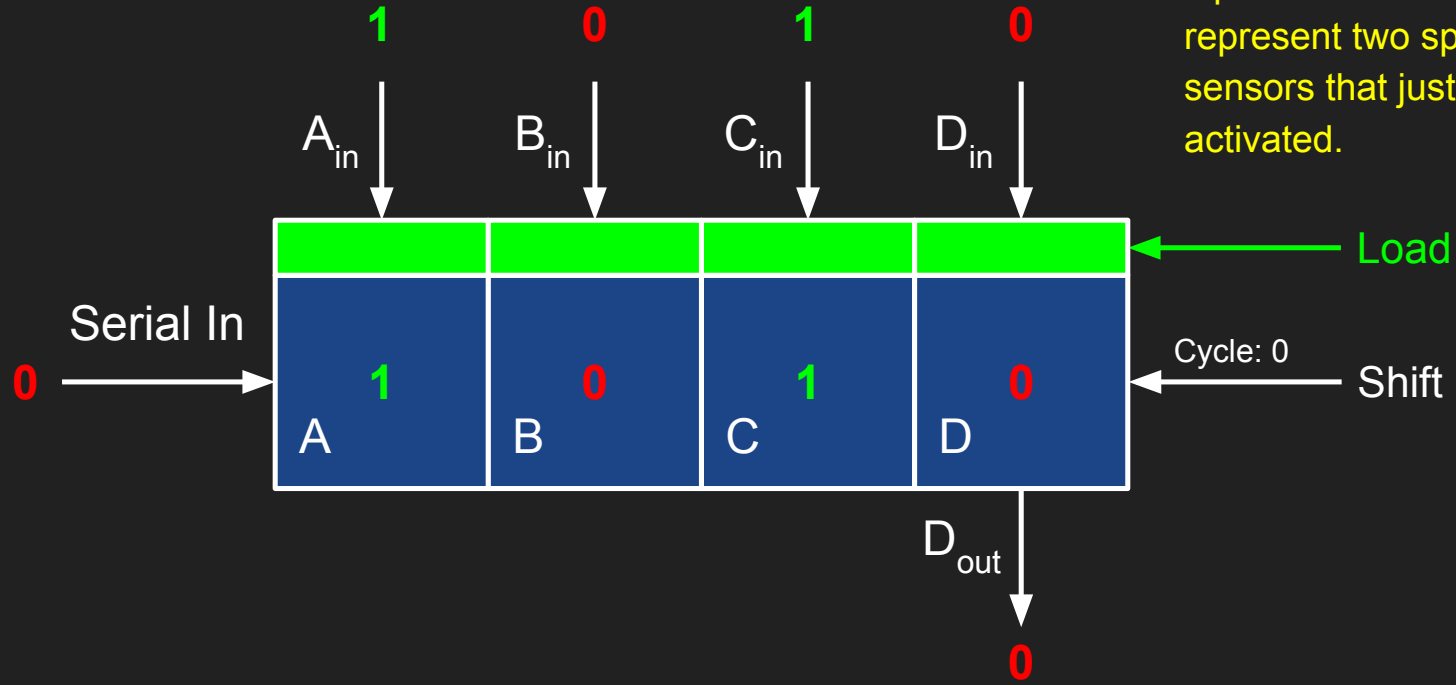


Shift register example

Let's say we just powered on the shift register and it initialized with all 0s. We also have some data on the input we want to read serially.

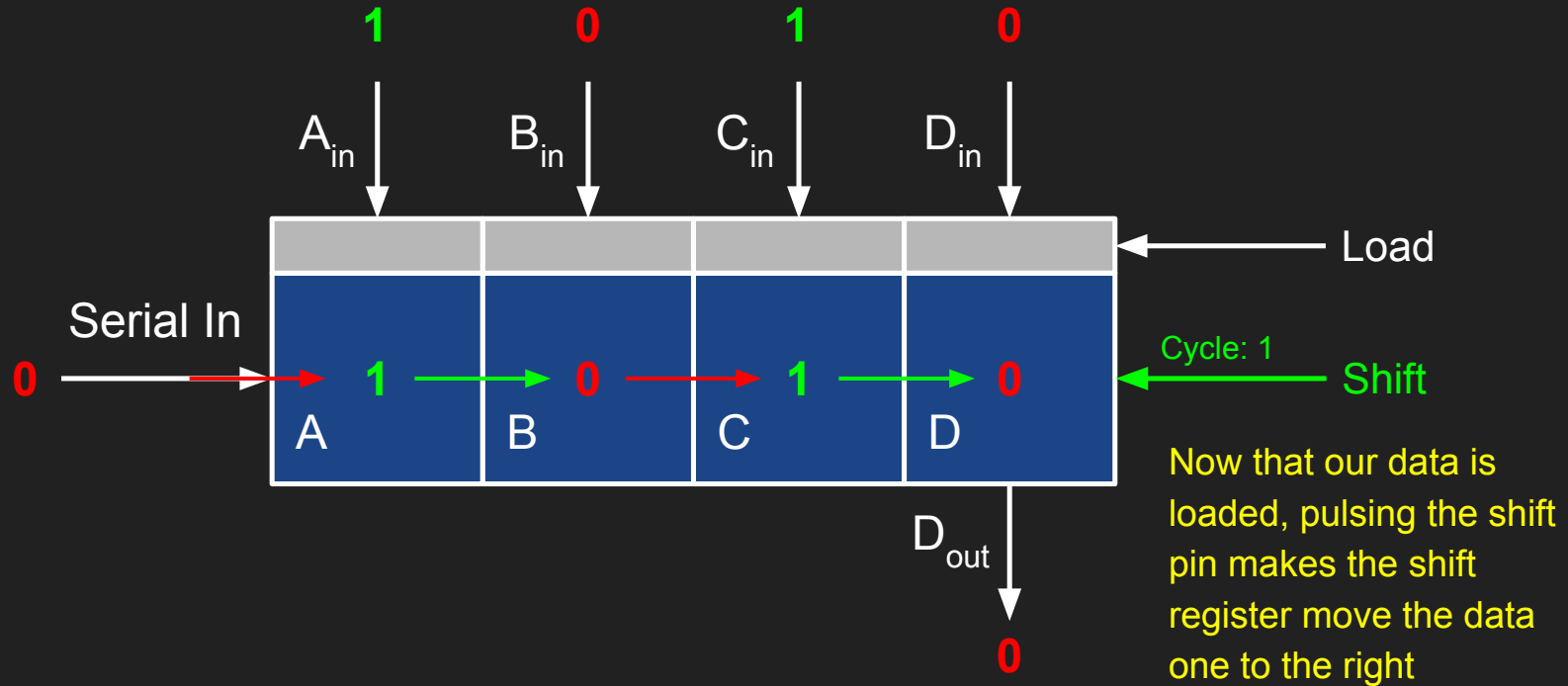


Shift register example

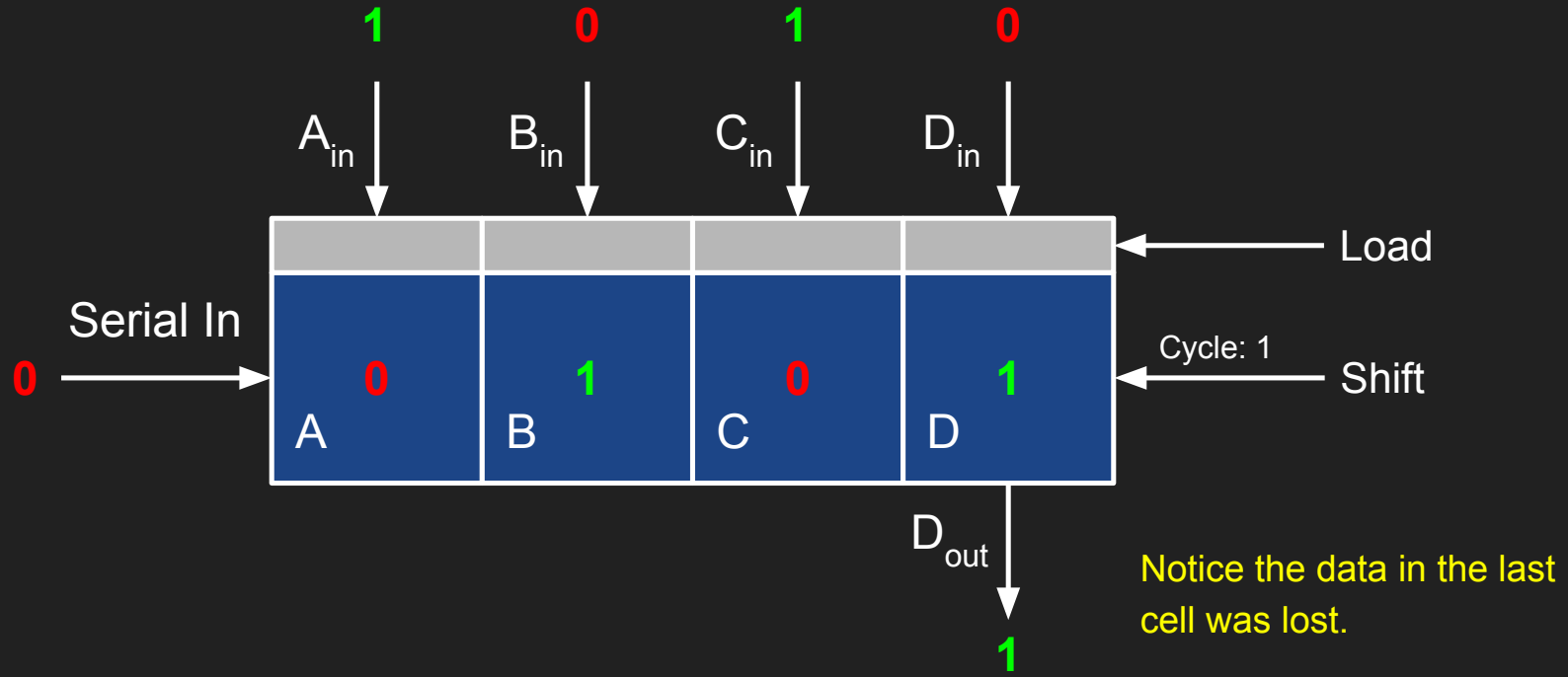


Pulsing the load pin makes the shift register store what it sees on the input. The two 1s maybe represent two speed sensors that just activated.

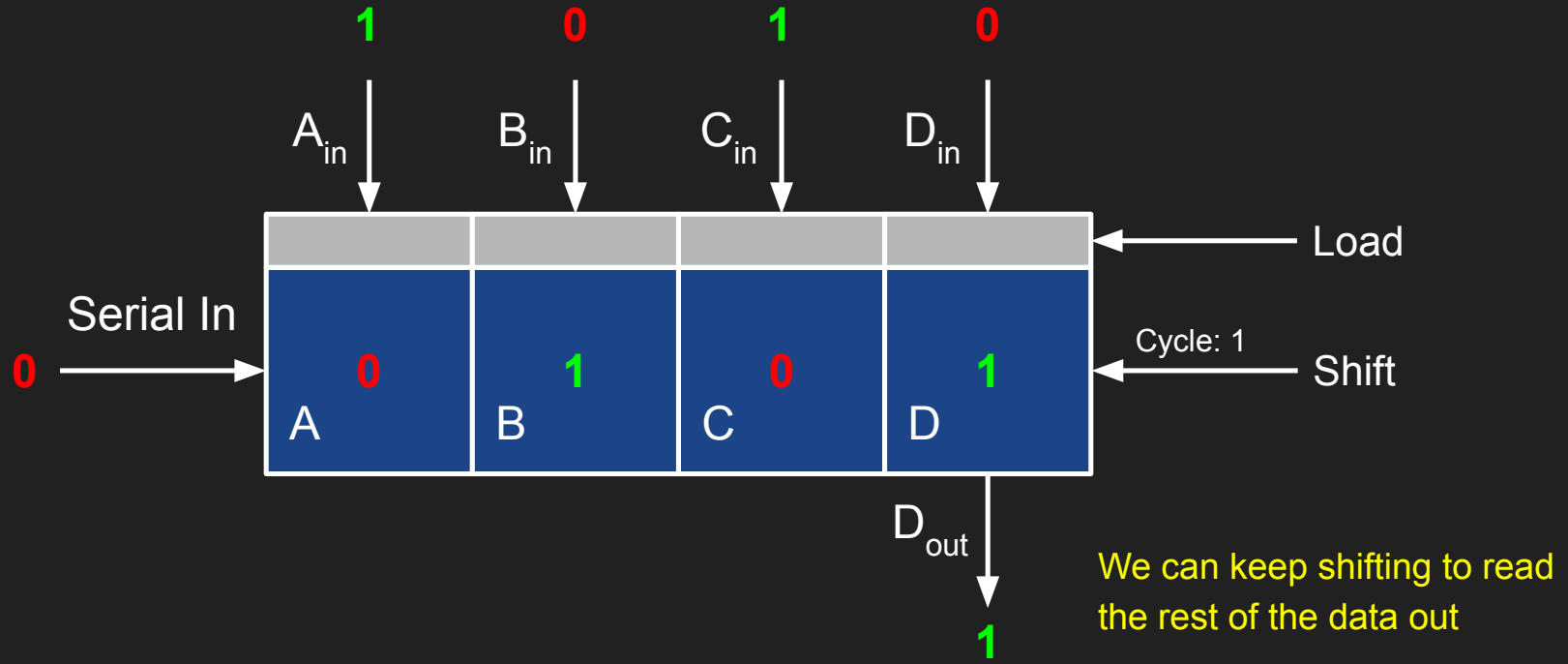
Shift register example



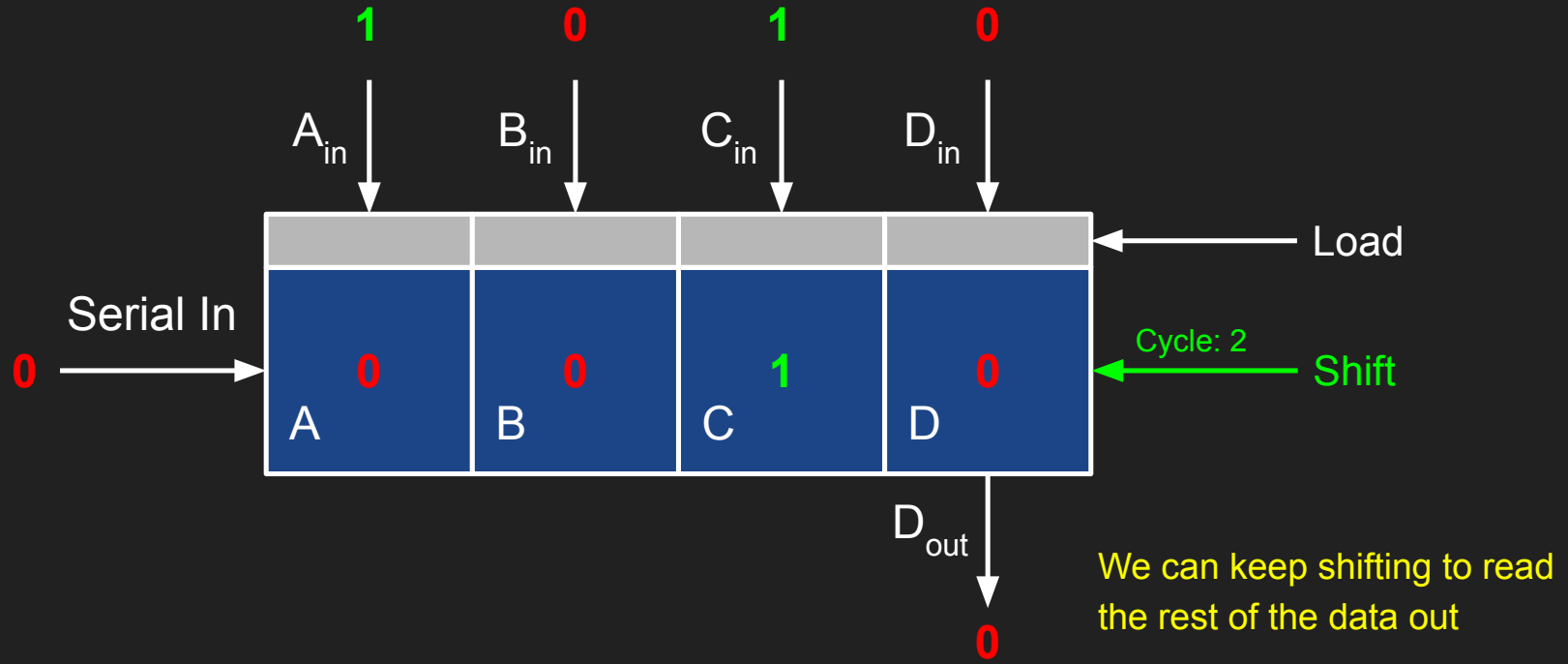
Shift register example



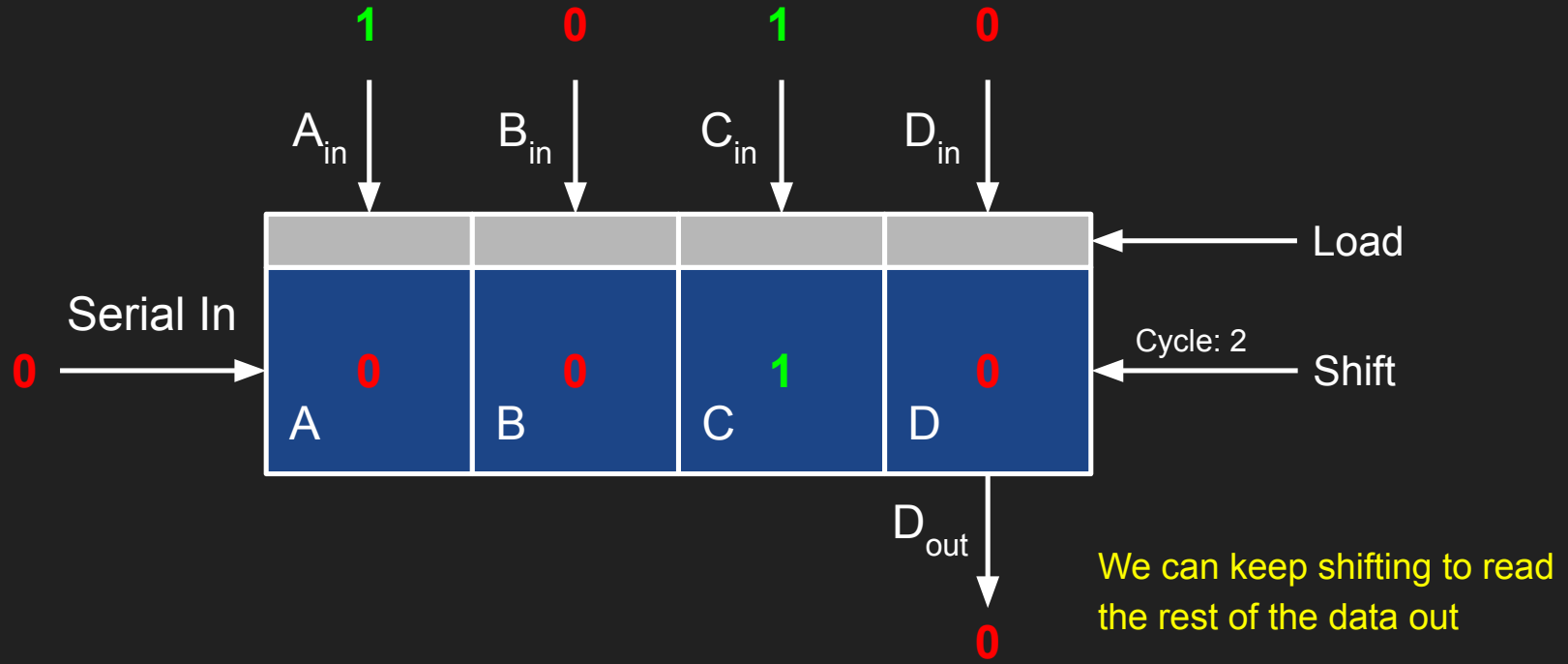
Shift register example



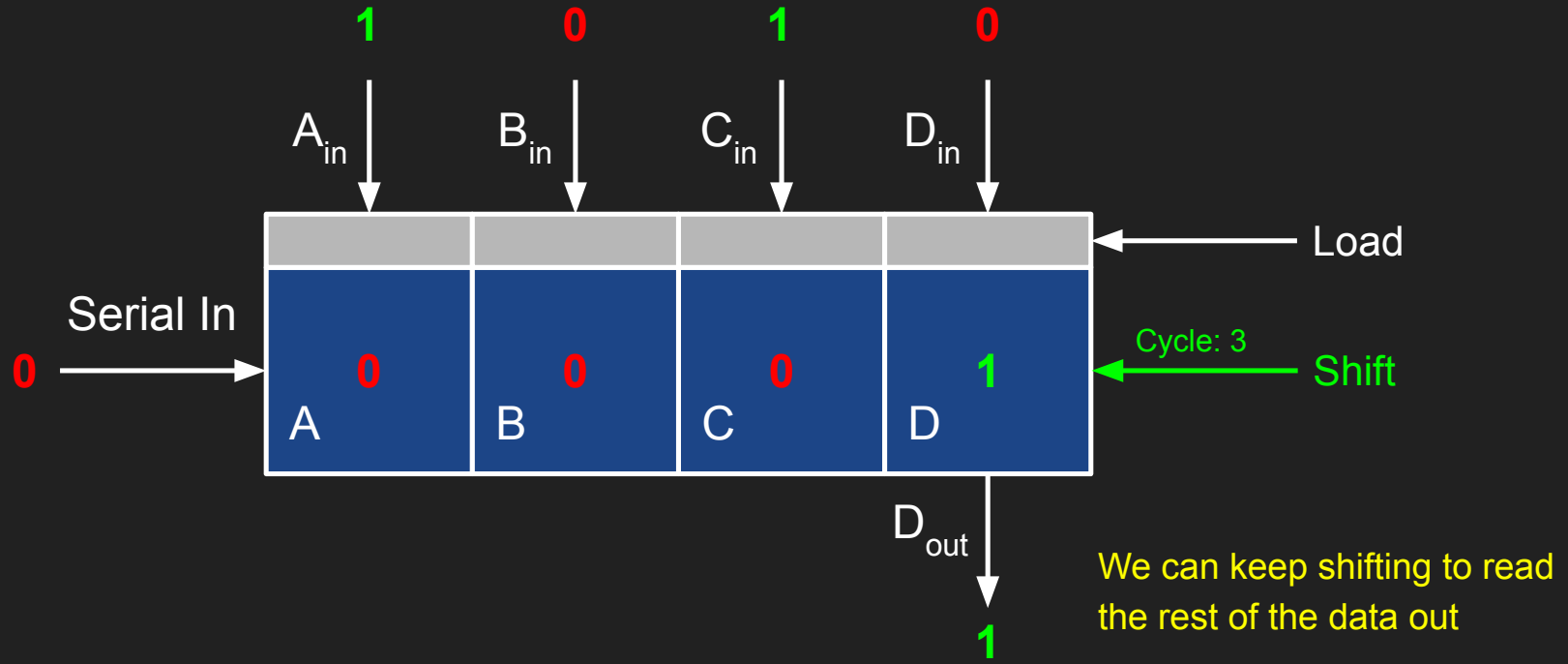
Shift register example



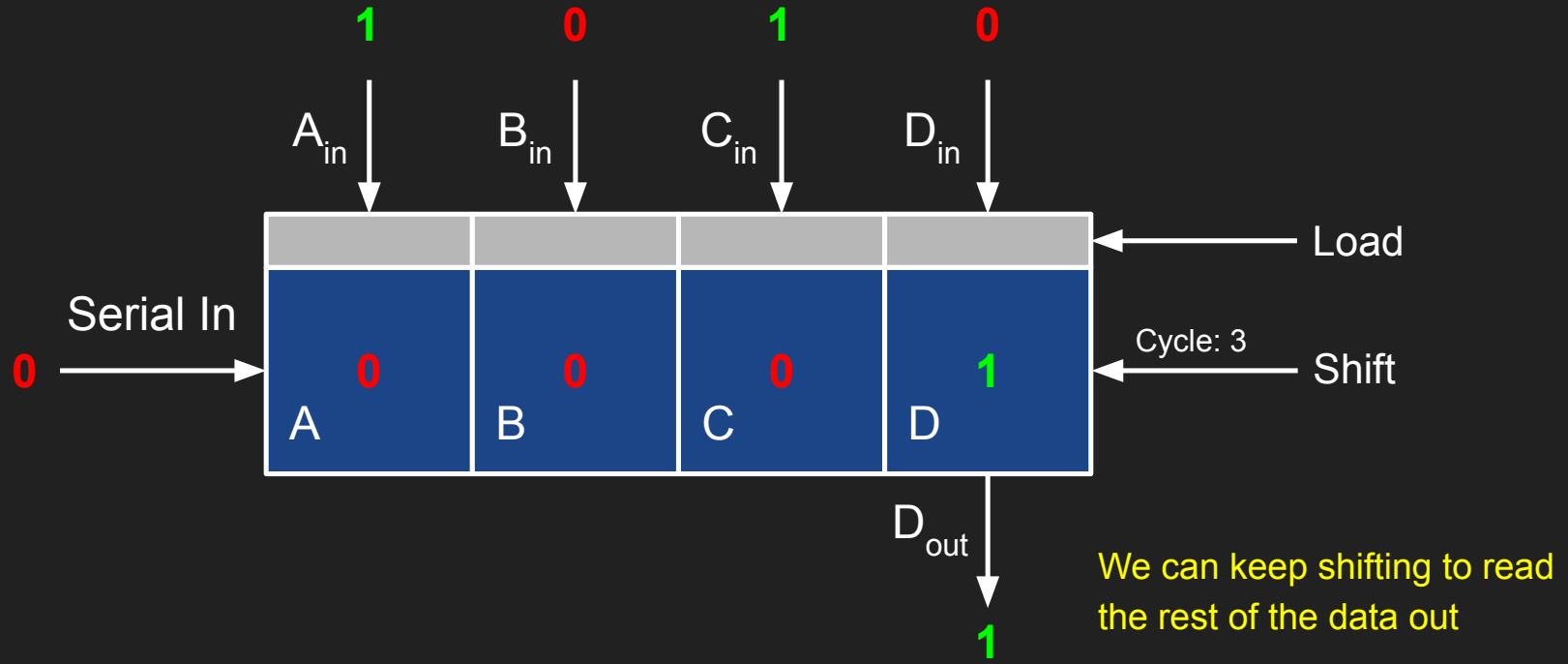
Shift register example



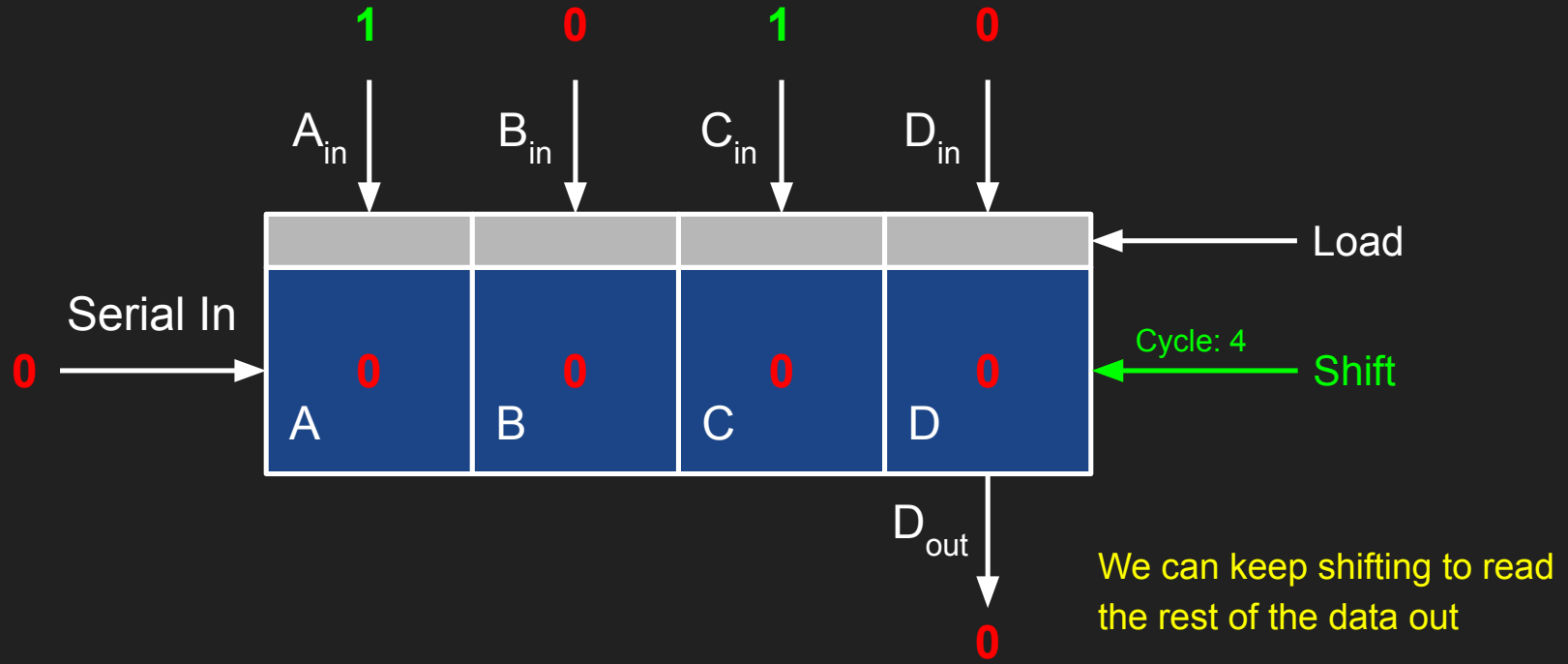
Shift register example



Shift register example

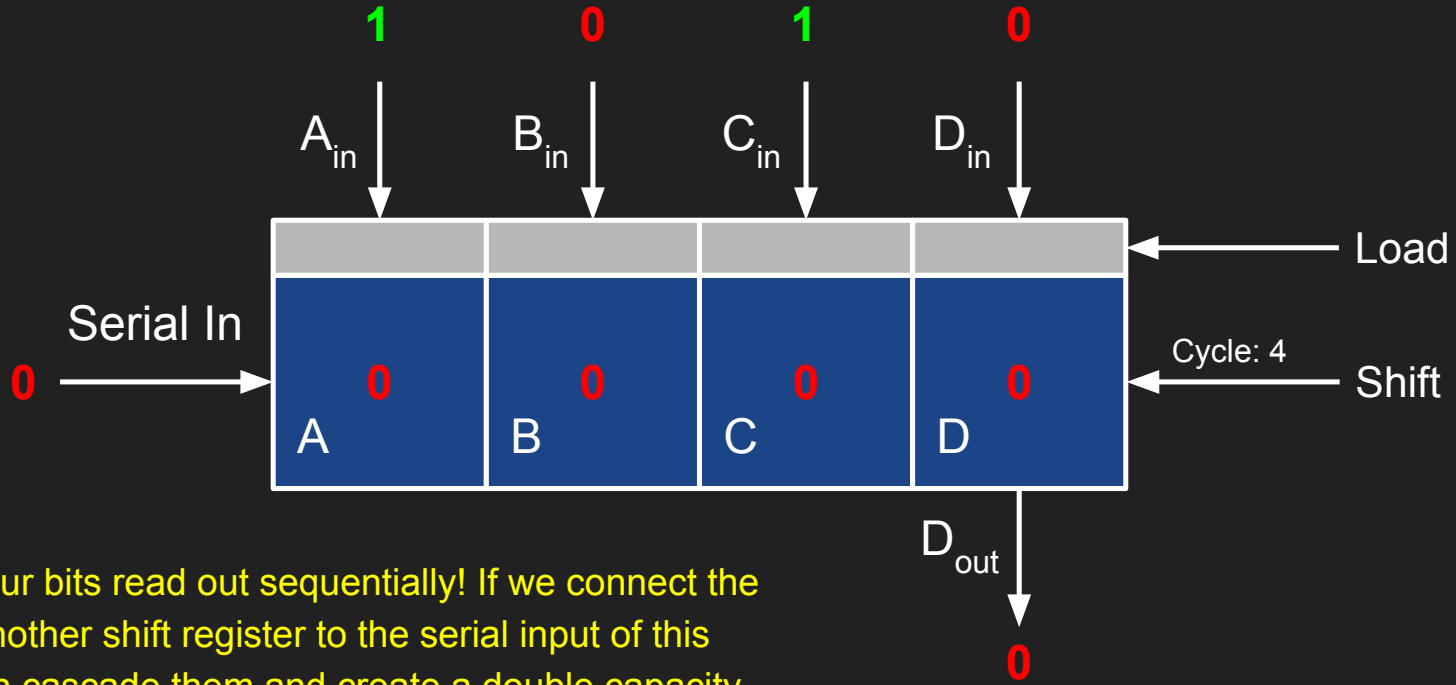


Shift register example



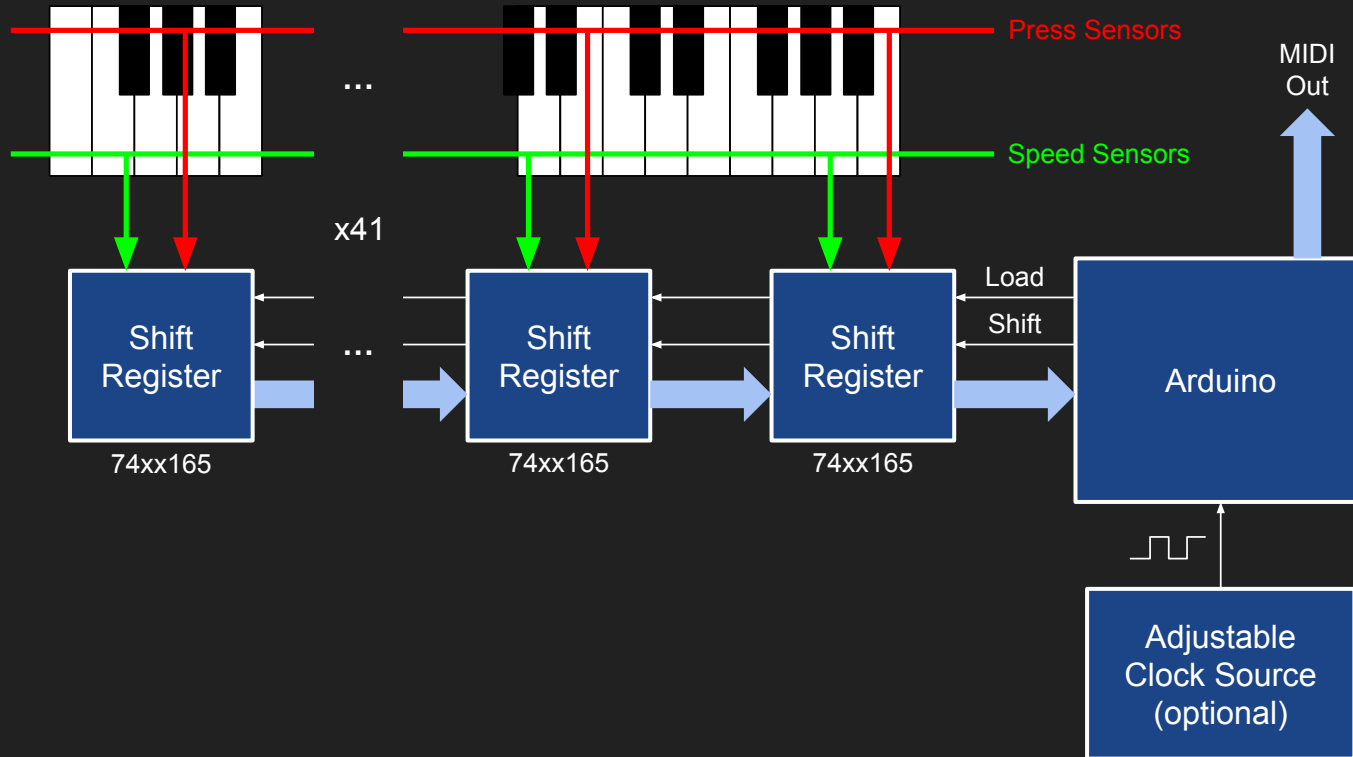
Shift register example

Notice how we don't care about what's on the input until we actually want to load new data in. This means we have to be careful to design the system so that we don't miss any changes on the input.



That's all four bits read out sequentially! If we connect the output of another shift register to the serial input of this one, we can cascade them and create a double capacity shift register.

Keyboard controller block diagram



- The 75xx165 shift register has 8 parallel inputs, meaning one chip per 4 keys, or 44 chips total
- Shift registers can be put in parallel for increased speed, i.e. one line for press and a parallel line for speed sensors clocked simultaneously

Arduino programming considerations

- Velocity timing should be done using the master timer accessible with the `millis()` or `micros()` functions subtracting start time from end time
- Add timeout feature to key sensing in case key is never pressed down all the way
- Investigate MIDI library for Arduino, writing our own would take too much time

Sound controller

Laptop?

